

Day 4 Deliverable — Dataset Design & Quality Expectations

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Dataset: LG HG2 18650 Li-ion Battery Dataset (McMaster University) | **Organization:** TIH IIT Guwahati | AI/ML Internship 2026

Part 1: Dataset Schema and Structure Note

1.1 Overview

The LG HG2 18650 Li-ion Battery Dataset was collected at McMaster University by Dr. Phillip Kollmeyer using a brand new 3Ah LG HG2 cylindrical lithium-ion cell tested on a 75A, 5V Digatron Battery Tester (0.1% voltage and current accuracy). The dataset spans 6 temperature conditions and contains 208 CSV files covering multiple test types.

1.2 Folder and File Structure

Files are organized by temperature into 6 folders:

Temperature Folder	No. of Files	Nominal Cell Temperature	Primary Use
40degC	35	40°C	High-temperature charging and discharge behavior
25degC	36	25°C	Room temperature baseline — largest file count
10degC	32	10°C	Mild cold temperature performance
0degC	32	0°C	Cold temperature — significant capacity reduction
n10degC	38	-10°C	Severe cold — charging acceptance drops significantly
n20degC	35	-20°C	Extreme cold — minimum capacity and highest internal resistance

1.3 File Types

Each temperature folder contains the following file types:

File Type	Count (all temps)	Description	Use in Project
Mixed1–8	46	Random combinations of UDDS, HWFET, LA92, US06 drive cycles	Primary training data for SoC model
Charge1–16	98	CC-CV charging sessions from different starting SoC levels	Primary data for Remaining Charging Time model
UDDS	6	Urban Dynamometer Driving Schedule — city driving profile	Test set (held out)
HWFET	6	Highway Fuel Economy Test — highway driving profile	Test set (held out)
LA92	6	LA92 Unified Cycle — aggressive city driving	Test set (held out)

File Type	Count (all temps)	Description	Use in Project
US06	6	US06 Supplemental cycle — high speed, aggressive acceleration	Test set (held out)
HPPC	6	Hybrid Pulse Power Characterization — pulse discharge for internal resistance characterization	Reference / not used in main model
Cap/Dis	28	Capacity and discharge tests at 0.5C, 1C, 2C rates	Reference for nominal capacity Q_nominal

1.4 Column Schema (per CSV file)

Each CSV file has 28 header rows of metadata followed by a row of column units and then the actual data. The 15 columns are:

Column Name	Unit	Data Type	Description
Time Stamp	MM/DD/YYYY HH:MM:SS	String → datetime	Timestamp at 0.1s resolution for drive cycles, slower for charge files
Step	Integer	Numeric	Test step number within the program sequence
Status	—	Categorical	TABLE (active measurement) or PAU (paused). Rows with PAU should be filtered out.
Prog Time	HH:MM:SS.ms	String → timedelta	Elapsed time since program start
Step Time	HH:MM:SS.ms	String → timedelta	Elapsed time within current step
Cycle	Integer	Numeric	Cycle count within the test
Cycle Level	Integer	Numeric	Sub-cycle level
Procedure	—	String	Name of the test program (e.g., LG_HG2_CyclesA)
Voltage	V	Float — KEEP	Cell terminal voltage. Range: 2.92V to 4.20V. Primary input feature.
Current	A	Float — KEEP	Positive = charge, Negative = discharge. Range: -16.89A to +6.0A. Primary input feature.
Temperature	°C	Float — KEEP	Battery case temperature. Range varies by folder (-20°C to 40°C). Primary input feature.
Capacity	Ah	Float — KEEP	Cumulative Ah counter. Resets to 0 after each test segment. Used to derive SoC labels.
WhAccu	Wh	Float	Cumulative energy counter. Resets after each segment. May be used as a derived feature.
Cnt	Integer	Numeric	Sample counter. Can be used to reconstruct sample index if timestamps are ambiguous.

1.5 Labels and Targets

Neither target variable is directly present in the raw files. Both must be derived:

- State of Charge (SoC) — Derived from the Capacity (Ah) column using Coulomb counting. Formula: $\text{SoC}(t) = \text{SoC_initial} + (\text{Capacity_t} / Q_nominal) \times 100$. $Q_nominal = 3.0$ Ah for LG HG2. The Capacity counter resets after each test segment, so SoC must be derived independently per segment, then stitched together where the initial SoC of each segment is known.
- Remaining Charging Time — Derived from the charging session files (Charge1–16). At each timestep during charging, Remaining Charging Time = $\text{timestamp_end_of_charge} - \text{timestamp_current}$. This requires identifying when charging is complete (current drops below $C/20 = 150\text{mA}$ threshold in CV phase).

1.6 Important Metadata

Cell model: LG HG2 18650 — NMC chemistry, 3Ah nominal capacity, 3.7V nominal voltage

Upper cutoff voltage: 4.2V (start of CV phase)

Lower cutoff voltage: 2.5V (end of discharge)

Charge current (CC phase): Approximately $0.5C = 1.5\text{A}$ for standard charging files

CV cutoff current: $C/20 = 150\text{mA}$ — charging ends when current drops below this threshold

Sample rate: 0.1s (10 Hz) for drive cycle files; slower (1-10s) for charging files

Tester accuracy: 0.1% for both voltage and current

Temperature sensor: AD592 at center of cell case — $\pm 1^\circ\text{C}$ accuracy

Battery condition: Brand new cell at start of testing

Part 2: Dataset Risk Checklist

The following risks have been identified and assessed before beginning data analysis. Each risk includes a mitigation plan.

2.1 Missing Data Risks

Risk	Likelihood	Impact	Mitigation
PAU status rows — tester paused, readings unreliable	Medium	Medium	Filter all rows where Status = PAU before any analysis or model training
NaN values in numeric columns from mixed-type parsing	Medium	Medium	Force numeric dtypes with <code>pd.to_numeric(errors='coerce')</code> . Inspect and document NaN counts per file.
Empty or near-empty charge files (130 rows only for some)	High	High	Exclude charge files with fewer than 500 rows from the Remaining Charging Time model. Document exclusions.
Missing timestamp entries due to tester communication dropout	Low	Medium	Check timestamp monotonicity per file. If gaps exist, interpolate or exclude affected segments.

2.2 SoC Derivation Risks

Risk	Likelihood	Impact	Mitigation
Capacity counter resets mid-session — SoC continuity breaks	High	High	Derive SoC per continuous segment only. Track reset points and stitch segments using known start SoC where available.
Unknown initial SoC at segment start	Medium	High	Use Cap_1C capacity test files to confirm Q_nominal. Assume 0% SoC at end of full discharge steps.
Coulomb counting drift over long sessions	Medium	Medium	Keep segment windows short. Cross-validate derived SoC against known full charge and discharge endpoints.
Q_nominal varies slightly with temperature	Medium	Medium	Use temperature-specific capacity from Cap files per temperature folder rather than a single fixed Q_nominal.

2.3 Data Leakage Risks

Risk	Likelihood	Impact	Mitigation
Random row shuffle before train/test split	High	High	NEVER shuffle rows. Always split chronologically by file or by time within a file.
Rolling window features spanning train/test boundary	Medium	High	Compute rolling features within each file separately. Never roll across file boundaries.
Future information in features (e.g., end-of-session SoC used mid-session)	Medium	High	Only use features available at the current timestep. No look-ahead features allowed.
Same drive cycle type appearing in both train and test sets	Medium	Medium	Train on Mixed1-6 files only. Reserve Mixed7-8 for validation and UDDS/HWFET/LA92/US06 for test.

2.4 Class / Target Imbalance Risks

Risk	Likelihood	Impact	Mitigation
SoC distribution skewed toward mid-range (20-80%) — drive cycles rarely reach extremes	High	Medium	Analyze SoC histogram before training. Evaluate model separately on low (0-20%), mid (20-80%), and high (80-100%) SoC ranges.
Remaining Charging Time skewed — most data from CV phase where time decreases slowly	Medium	Medium	Analyze charging time distribution. Consider stratified sampling if CC phase is underrepresented.
File count imbalance across temperatures (n10degC has 38, 0degC and 10degC have only 32)	Low	Low	Note imbalance but do not artificially balance. Evaluate per-temperature performance separately.

2.5 Temporal and Annotation Risks

Risk	Likelihood	Impact	Mitigation
Variable sample rates across file types (0.1s for drive cycles, slower for charge files)	High	Medium	Do not assume uniform timestep. Use timestamps directly for time-based features. Resample if needed.

Risk	Likelihood	Impact	Mitigation
Temperature in drive cycle files may not match folder temperature — battery self-heats under load	Medium	Low	Use actual measured temperature column, not folder name, as the temperature feature.
Remaining Charging Time label ambiguity — when exactly is charging 'complete'?	Medium	Medium	Define completion as: current drops below 150mA (C/20) threshold in CV phase. Document and apply consistently.

2.6 Train / Validation / Test Split Definition

The following split strategy is defined to preserve temporal order, prevent leakage, and ensure the test set reflects real-world unseen conditions:

Split	Files Included	Rationale
Training	Mixed1, Mixed2, Mixed3, Mixed4, Mixed5, Mixed6 from all 6 temperature folders + Charge1–12	Random mixed drive cycles cover diverse SoC trajectories. Charge files provide CC-CV phase data for charging time target.
Validation	Mixed7 from all 6 temperature folders + Charge13–14	Same cycle type as training but different session. Used for hyperparameter tuning and early stopping.
Test	Mixed8, UDDS, HWFET, LA92, US06 from all 6 temperatures + Charge15–16	Standard drive cycles are completely unseen during training. Tests generalization to real-world EV usage patterns.

Split rules: (1) Never shuffle rows within a file. (2) Never split a single file across train and test — keep each file entirely in one split. (3) Rolling window features must be computed within each file before splitting. (4) Scaler (StandardScaler) must be fit on training data only and applied to validation and test.